INCH-POUND
MIL-DTL-24643/7G
27 November 2023
SUPERSEDING
MIL-DTL-24643/7F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LSMHOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First	-	Copper conductor, coated or uncoated, 16 AWG, class K stranding.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Minimum average wall thickness of 0.020 inch. Standard identification code applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.
Fourth	-	The specified number of conductors (see table I) shall be cabled together as required for the completed cable to meet the twisting and bending endurance tests. The outer layer shall be a left-hand lay. The inner layer(s) lay direction at the manufacturer's option. Fillers may be used as necessary to form a firm, well-rounded assembly. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. A single conductor shall not be used in the center for the cable core.
Fifth	-	An optional binder.
Sixth	-	Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.060 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/7	Type and size	Number of conductors	Overall diameter (inches)		Roller diameter for bending endurance max. (inches)
			min.	max.	
-01UN	LSMHOF-7	7	0.465	0.500	2.50
-02UN	LSMHOF-10	10	0.540	0.585	2.50
-03UN	LSMHOF-14	14	0.585	0.635	3.00
-04UN	LSMHOF-19	19	0.650	0.705	3.00
-05UN	LSMHOF-24	24	0.735	0.795	3.50
-06UN	LSMHOF-30	30	0.775	0.835	4.00
-07UN	LSMHOF-37	37	0.855	0.925	4.00
-08UN	LSMHOF-44	44	0.925	1.000	4.50
-09UN	LSMHOF-61	61	1.100	1.175	5.00

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	4.59
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Twisting and bending endurance (see table I for roller size)	
At 167 °F (75 °C) (cycles, min.)	6,000
At -22±3.6 °F (-30±2 °C) (cycles, min.)	2,000

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Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18

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Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSMHOF (all sizes)	1,500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-023)

Review activities:
Army – AR,CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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